Inference for categorical data

Getting Started

```
library(tidyverse)
library(openintro)
library(infer)


yrbss |>
  count(text_while_driving_30d)
```

```
# A tibble: 9 x 2
  text_while_driving_30d     n
  <chr>                  <int>
1 0                       4792
2 1-2                      925
3 10-19                    373
4 20-29                    298
5 3-5                      493
6 30                       827
7 6-9                      311
8 did not drive           4646
9 <NA>                     918
```

Exercise 1

```
yrbss |>
  count(text_while_driving_30d)
```

```
# A tibble: 9 x 2
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6 30                       827
7 6-9                      311
8 did not drive           4646
9 <NA>                     918
```

Exercise 2

```
no_helmet_text_daily <- yrbss %>%
  filter(helmet_12m == "never") |>
  filter(text_while_driving_30d == "30")

no_helmet_txters <- no_helmet_text_daily |>
  count()

total_surveyed <- yrbss|>
  count()

proportion <- no_helmet_txters[[1]] / total_surveyed[[1]]
```

The total proportion of people who texted while driving everyday and have never worn helmets is 0.0340867

Inference on proportions

```
data('yrbss', package='openintro')
no_helmet <- yrbss %>%
  filter(helmet_12m == "never")

no_helmet <- no_helmet %>%
  mutate(text_ind = ifelse(text_while_driving_30d == "30", "yes", "no"))
```

```r
boot_dist <- no_helmet %>%
  filter(!is.na(text_ind)) %>%
  specify(response = text_ind, success = "yes") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "prop") %>%
  get_ci(level = 0.95)
```

Exercise 3

```r
boot_dist <- no_helmet |>
  filter(!is.na(text_ind)) |>
  specify(response = text_ind, success = "yes") |>
  generate(reps = 1000, type = "bootstrap") |>
  calculate(stat = "prop")

ci <- boot_dist |>
  get_ci(level = 0.95, type = "percentile")
ci
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
     <dbl>    <dbl>
1   0.0649   0.0780
```

```r
margin <- (ci$upper_ci - ci$lower_ci) / 2
```

The margin for error is 0.0065354

Exercise 4

```r
gender_data <- yrbss |>
  filter(!is.na(gender))

boot_gender <- gender_data |>
  specify(response = gender, success = "female") |>
  generate(reps = 1000, type = "bootstrap") |>
  calculate(stat = "prop")
```

```r
ci_gender <- boot_gender |>
  get_ci(level = 0.95, type = "percentile")
ci_gender
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
     <dbl>    <dbl>
1    0.479    0.496
```

```r
(ci_gender$upper_ci - ci_gender$lower_ci) / 2
```

```
[1] 0.008622246
```

```r
active_data <- yrbss |>
  filter(!is.na(physically_active_7d)) |>
  mutate(active_all_week = ifelse(physically_active_7d == 7, "yes", "no"))

boot_active <- active_data |>
  specify(response = active_all_week, success = "yes") |>
  generate(reps = 1000, type = "bootstrap") |>
  calculate(stat = "prop")

ci_active <- boot_active |>
  get_ci(level = 0.95, type = "percentile")
ci_active
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
     <dbl>    <dbl>
1    0.265    0.279
```

```r
(ci_active$upper_ci - ci_active$lower_ci) / 2
```

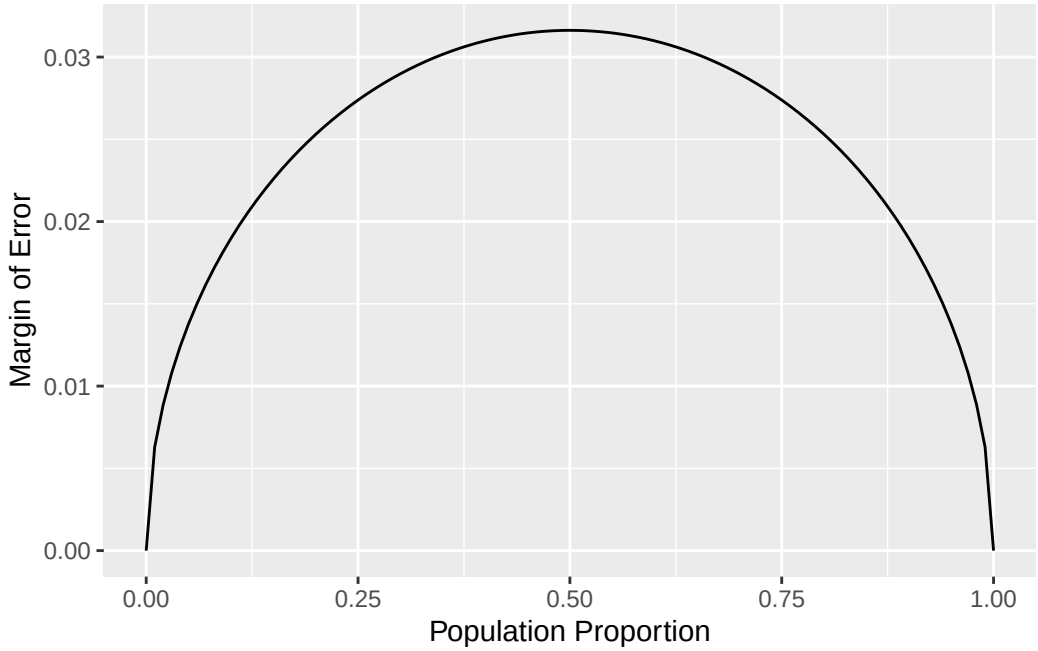
```
[1] 0.007062359
```

Proportion and margin for error

```r
n <- 1000

p <- seq(from = 0, to = 1, by = 0.01)
me <- 2 * sqrt(p * (1 - p)/n)

dd <- data.frame(p = p, me = me)
ggplot(data = dd, aes(x = p, y = me)) +
  geom_line() +
  labs(x = "Population Proportion", y = "Margin of Error")
```




```r
.1 + 2*(sqrt(0.1 * 0.9 / 300))
```

```
[1] 0.134641
```

The relationship between p and me based on the graph is a symmetric upside down arch centered at .5

Success-failure conditions

Exercise 6

Center: ~.1

Spread: between 0.134641 and 0.065359

Shape: about normal

Exercise 7

The center moves with p.

The spread is largest near p = .5 and smaller as p approaches 0 or 1.

shape is symmetric when p = .5 and skews as it approaches 0 or 1

Exercise 8

The center doesn't change.

Spread increase with a larger n

Shape becomes more symmetric and normal as n increases

Exercise 9

```
yrbss_sleep <- yrbss |>
  filter(!is.na(school_night_hours_sleep), !is.na(strength_training_7d)) |>
  mutate(sleep_10 = ifelse(school_night_hours_sleep == "10+", "10+", "less"),
         st_daily = ifelse(strength_training_7d == 7, "yes", "no"))

obs_diff <- yrbss_sleep |>
  specify(st_daily ~ sleep_10, success = "yes") |>
  calculate(stat = "diff in props", order = c("10+", "less"))

null_dist <- yrbss_sleep |>
  specify(st_daily ~ sleep_10, success = "yes") |>
  hypothesize(null = "independence") |>
  generate(reps = 1000, type = "permute") |>
  calculate(stat = "diff in props", order = c("10+", "less"))
```

```
get_p_value(null_dist, obs_stat = obs_diff, direction = "greater")
```

```
Warning: Please be cautious in reporting a p-value of 0. This result is an approximation
based on the number of `reps` chosen in the `generate()` step.
i See `get_p_value()` (`?infer::get_p_value()`) for more information.
```

```
# A tibble: 1 x 1
  p_value
    <dbl>
1       0
```

```
yrbss_sleep |>
  specify(st_daily ~ sleep_10, success = "yes") |>
  generate(reps = 1000, type = "bootstrap") |>
  calculate(stat = "diff in props", order = c("10+", "less")) |>
  get_ci(level = 0.95)
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
     <dbl>    <dbl>
1   0.0582    0.154
```

```
(1.95/.01)^2 * .25
```

```
[1] 9506.25
```

according to the results we can determine that those that sleep more than 10 hours a night are more likely to strength train every day

Exercise 10

One would expect to detect one by chance about 5% of the time.

Exercise 11

with no idea what p is, we would use .5 as the worst case scenario. Then solve for n in 1.96 * sqrt(.5*.5/n) <= .01 which is n = 9506.25